

Nonemptiness of the Clifford Spectrum for Three Hermitian Matrices of Arbitrary Finite Size

Candidate solution packet for arXiv:2602.15302

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Status. Candidate full solution, likely valid pending expert review. The proof gives a quantitative strengthening of the arbitrary-size, three-matrix conjecture highlighted in the source paper.

1 Source problem and scope

Cerjan, Lauric, and Loring prove that the Clifford spectrum of any three 2×2 Hermitian matrices is nonempty [1]. Their abstract describes a conjecture for three $n \times n$ matrices of arbitrary size, while the introduction says that general nonemptiness of the Clifford spectrum remains open and calls their 2×2 theorem “the smallest nontrivial case” (source PDF, page 3). The earlier paper of Cerjan and Loring records the broader all-tuples conjecture explicitly [2, Conjecture 8.1].

The relevant passage from page 3 of the source paper is reproduced below.

For many forms joint spectrum for noncommuting matrices [18, 25] we find that we do not know that the joint spectrum must always be nonempty, and indeed sometimes this is false [8]. The Clifford spectrum, where $L_{\lambda}(\mathbf{A})$ is not just close to singular, but is actually singular, is critical for the applications to physics as these singularities correspond to changes in local topology in a material. The many calculations done by physicists using the spectral localizer amount to ample evidence that the Clifford spectrum is never empty.

Kisil [20] claimed to have shown the Clifford spectrum is always nonempty by showing that the related monogenic spectrum is nonempty. It was subsequently shown [19] that these spectra are not always equal, so the conjecture on the Clifford spectrum is still open. The Clifford spectrum is the only joint spectrum known to have applications to topological protection in materials so this conjecture seems important.

Here, we give a complete answer in the smallest nontrivial case: for any three 2×2 Hermitian matrices, the Clifford spectrum is never empty. After reducing by unitary conjugacy so that A_1 is diagonal and expanding the localizer in the Pauli basis, we derive a closed formula for $\det[L_{\lambda}(\mathbf{A})]$ that is a quartic polynomial $D(\lambda)$ with an explicit symmetry center and then show that $D(\lambda) \leq 0$ for some $\lambda \in \mathbb{R}^3$. Equivalently, $L_{\lambda}(\mathbf{A})$ is singular somewhere for every triple $\mathbf{A}$, establishing non-emptiness of their Clifford spectrum.

For the three-matrix case, fix the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Given Hermitian matrices $A_1, A_2, A_3 \in M_n(\mathbb{C})$ and $\lambda = (\lambda_1, \lambda_2, \lambda_3) \in \mathbb{R}^3$, their localizer is

$$L_\lambda(A) = \sum_{j=1}^3 (A_j - \lambda_j I_n) \otimes \sigma_j.$$

The standard Pauli Clifford spectrum is

$$\Lambda^C(A) = \{\lambda \in \mathbb{R}^3 : L_\lambda(A) \text{ is singular}\}.$$

The source problem in the present scope asks whether this set is nonempty for every finite n , not only for $n = 2$.

2 Proof intuition

Far from the matrices, the localizer is governed by the universal Pauli term $-\sum_j \lambda_j I_n \otimes \sigma_j$. On a large sphere this term has a uniform spectral gap, so the actual localizer can be deformed to it without crossing zero. The negative eigenspaces of the universal term form n copies of the Hopf line bundle over S^2 . Their determinant has winding number $\pm n$ and is therefore nontrivial.

If the localizer had no singularity inside the sphere, however, its negative spectral subspaces would make a vector bundle over the whole ball. Every bundle over a ball is trivial, so its determinant would restrict to a trivial line bundle on the boundary. The nonzero Hopf winding gives the contradiction. The size of the universal gap also locates a singularity in a closed ball determined by the norm of the localizer at the origin.

3 The theorem

Theorem 1 (Arbitrary finite size, with a norm bound). *Let $n \geq 1$ and let $A_1, A_2, A_3 \in M_n(\mathbb{C})$ be Hermitian. Set*

$$K = \sum_{j=1}^3 A_j \otimes \sigma_j, \quad C = \|K\|.$$

Then there is a $\lambda \in \mathbb{R}^3$ such that

$$\|\lambda\|_2 \leq C \quad \text{and} \quad L_\lambda(A) \text{ is singular.}$$

In particular, $\Lambda^C(A)$ is nonempty for every finite n .

Lemma 1 (The boundary Hopf class). *For $\omega \in S^2$, let $H_+(\omega)$ be the $+1$ eigenspace of $\omega \cdot \sigma = \sum_{j=1}^3 \omega_j \sigma_j$. The line bundle H_+ over S^2 has a transition function of winding number -1 . Consequently, the determinant line of $H_+^{\oplus n}$ has winding number $-n$ and is nontrivial.*

Proof. Write

$$\omega = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

On the complement of the south pole and the complement of the north pole, respectively, normalized $+1$ eigenvectors are

$$u_N(\theta, \phi) = \begin{pmatrix} \cos(\theta/2) \\ e^{i\phi} \sin(\theta/2) \end{pmatrix}, \quad u_S(\theta, \phi) = \begin{pmatrix} e^{-i\phi} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix}.$$

Direct multiplication verifies $(\omega \cdot \sigma)u_N = u_N$ and $(\omega \cdot \sigma)u_S = u_S$. On the overlap, $u_S = e^{-i\phi}u_N$. Thus the equatorial transition function is $e^{-i\phi}$, of winding -1 . For n direct summands the transition matrix is $e^{-i\phi}I_n$, whose determinant is $e^{-in\phi}$. Its nonzero winding proves that the determinant line, and hence the vector bundle, is nontrivial. $\square$

Proof of Theorem 1. If $C = 0$, then $L_0(A) = K = 0$ and the assertion holds with $\lambda = 0$. Assume henceforth that $C > 0$.

Fix any $R > C$. We first prove that $L_\lambda(A)$ is singular at some point of the closed ball $B_R = \{\lambda : \|\lambda\|_2 \leq R\}$. Suppose, to the contrary, that it is invertible everywhere on B_R . Because the localizer is Hermitian and depends continuously on λ , the negative spectral projections

$$P(\lambda) = \mathbf{1}_{(-\infty, 0)}(L_\lambda(A))$$

vary continuously and define a complex vector bundle $E \rightarrow B_R$.

For $\omega \in S^2$ and $t \in [0, 1]$, consider the boundary homotopy

$$F_t(\omega) = tK - RI_n \otimes (\omega \cdot \sigma).$$

Its endpoint at $t = 1$ is the actual boundary localizer $L_{R\omega}(A)$. Since $(\omega \cdot \sigma)^2 = I_2$, the operator $RI_n \otimes (\omega \cdot \sigma)$ has inverse norm $1/R$. More directly, for every vector v ,

$$\begin{aligned} \|F_t(\omega)v\| &\geq R\|v\| - t\|Kv\| \\ &\geq (R - tC)\|v\| \geq (R - C)\|v\| > 0. \end{aligned}$$

Thus $F_t(\omega)$ is invertible throughout $S^2 \times [0, 1]$.

At $t = 0$, the negative eigenspace of $F_0(\omega) = -RI_n \otimes (\omega \cdot \sigma)$ is

$$\mathbb{C}^n \otimes H_+(\omega) \cong H_+(\omega)^{\oplus n}.$$

It has rank n . The negative spectral projections of F_t give a homotopy of boundary bundles, so

$$E|_{\partial B_R} \cong H_+^{\oplus n}$$

up to bundle homotopy; in particular their determinant lines have the same winding number. Lemma 1 says this number is $-n \neq 0$.

On the other hand, B_R is contractible, so the vector bundle $E \rightarrow B_R$ is trivial. Its determinant line is therefore trivial, and its restriction to $\partial B_R \cong S^2$ must have winding number zero. This contradicts the preceding paragraph. Hence every $R > C$ contains a point λ_R at which $L_{\lambda_R}(A)$ is singular.

Choose $R_k = C + 1/k$. The points λ_{R_k} lie in the fixed compact ball B_{C+1} , so a subsequence converges to some λ with $\|\lambda\|_2 \leq C$. Since the determinant is continuous,

$$\det L_\lambda(A) = \lim_{k \rightarrow \infty} \det L_{\lambda_{R_k}}(A) = 0.$$

This proves the theorem. $\square$

Remark 1 (Other three-generator representations). *Every finite-dimensional complex representation of the three-generator Clifford relations is a direct sum of irreducible chiral representations. An orientation reversal changes the sign of the Hopf winding but not its nonvanishing. Hence each irreducible summand has a nonempty spectrum by the same proof, and a direct-sum localizer is singular wherever one summand is. The theorem was stated with the source paper's Pauli convention to keep the scope exact.*

4 Verification and limitations

The proof is finite-dimensional and does not address triples of unbounded or infinite-dimensional self-adjoint operators. It also does not settle the broader conjecture for an arbitrary number of matrices; the obstruction used here is specifically the nontrivial boundary class of the Pauli map on S^2 .

The boundary gap, negative rank, explicit transition function, extension contradiction, and limiting norm bound were audited separately in `VERIFICATION.md`. The optional script `code/verify_boundary_obstruction.py` checks the gap and winding on random finite examples, but no numerical statement is used in the proof.

5 Novelty check

A bounded search on 9 August 2026 covered the run’s four lightweight indexes, the exact arXiv identifier and title, and arXiv/web combinations of “Clifford spectrum” with “Hopf line bundle,” “negative spectral subspace,” “Chern class,” “Pauli,” and “nonempty.” It found the source 2×2 theorem [1] and the earlier general conjecture [2], but no proof of the arbitrary- n three-matrix case and no appearance of the argument above. Novelty confidence is therefore moderate, not definitive, pending specialist review and broader citation-database searches.

6 Human-review recommendation

The packet is ready for expert review as a candidate full solution. The highest-value checks are the match between the source’s intended Clifford representation and the Pauli convention, and the standard implication that the boundary negative bundle induced by an invertible filling must have trivial determinant class. The matrix estimate and Hopf transition calculation are explicit.

References

- [1] A. Cerjan, V. Lauric, and T. A. Loring, *Clifford spectrum of three 2 by 2 matrices*, arXiv:2602.15302v1, 2026.
- [2] A. Cerjan and T. A. Loring, *Even spheres as joint spectra of matrix models*, J. Math. Anal. Appl. **531** (2024), 127892; arXiv:2305.12026v2.